# AE700 Detailed Study Material
# Lectures 18 to 23

Prepared from course lectures: Guidance and Control of Unmanned Autonomous Vehicles

April 27, 2026

## Contents


**1 Lecture 18: TTC via Flat-Earth Model, Precision Landing, and Multirotor Introduction 3**

1.1 Time-to-Collision using Flat-Earth Model . . . 3

1.2 Precision Landing via Proportional Navigation Guidance . . . 3

1.3 Fixed-wing vs. Multirotor UAVs . . . 4

1.4 Quadrotor Movements (Physical Intuition) . . . 5

**2 Lecture 19: Quadrotor Hardware and Modelling Architecture 6**

2.1 Structural Components . . . 6

2.2 Airframe Configurations . . . 6

2.3 CG Location . . . 7

2.4 Autopilot Placement . . . 7

2.5 Aerodynamic Drag on Fuselage . . . 7

2.6 Modelling Architecture . . . 8

**3 Lecture 20: Quadrotor Kinematics and Dynamics 9**

3.1 Coordinate Frames and Euler Angles . . . 9

3.2 Rigid-Body Kinematics . . . 9

3.3 Rigid-Body Newton–Euler Dynamics . . . 9

3.4 Quadrotor Force and Moment Model . . . 10

3.5 Propulsor Model . . . 10

3.6 Blade Flapping and Hub Forces . . . 10

**4 Lecture 21: Quadrotor Position and Velocity Control 11**

4.1 Cascaded Control Architecture . . . 11

4.2 Position Dynamics . . . 11

4.3 Altitude (Z) Controller . . . 11

4.4 Horizontal (X–Y) Controller . . . 11
4.5 Attitude Reference Generation . . . 12

**5 Lecture 22: Quadrotor Attitude Control, Control Allocation, and Motor Control 13**

5.1 Attitude Control Objectives . . . 13
5.2 Euler Angle Based Attitude Control . . . 13
5.3 Rotation Matrix (SO(3)) Based Attitude Control . . . 13
5.4 Control Allocation . . . 14
5.5 Motor Speed Control . . . 15
5.6 Simulation Parameters (Example) . . . 15

**6 Lecture 23: Fully Actuated Unmanned Autonomous Vehicles 16**

6.1 Motivation . . . 16
6.2 Classification: Fixed-tilt vs. Variable-tilt . . . 16
6.3 Coordinate Frames and Propeller Configuration . . . 16
6.4 Aerodynamic Wrench and Control-Wrench Matrix . . . 16
6.5 Actuation Classification . . . 17
6.6 Fixed-tilt FA-UAV Designs . . . 17
6.7 Variable-tilt FA-UAV Designs . . . 19
6.8 Summary Table: FA-UAV Configurations . . . 20

**7 Consolidated Revision Checklist (Lectures 18–23) 21**

# 1 Lecture 18: TTC via Flat-Earth Model, Precision Landing, and Multirotor Introduction

## 1.1 Time-to-Collision using Flat-Earth Model

In Lecture 17, TTC required knowledge of the object's physical size. Here a flat-earth assumption removes that requirement.

**Flat-earth geometry.** For a camera looking downward at angle $\varphi$ from vertical, and UAV height $h = -p_d$:

$$\mathbb{L} = \frac{h}{\cos\varphi}, \qquad h = -p_d.$$

The $z$-component of the unit inertial LOS gives $\cos\varphi = \tilde{l}^i_z$.

**Differentiation for TTC.** Differentiating:

$$\frac{\dot{\mathbb{L}}}{\mathbb{L}} = \frac{1}{\mathbb{L}}\frac{d}{dt}\left(\frac{h}{\cos\varphi}\right) = \frac{\dot{h}}{h} + \dot{\varphi}\tan\varphi.$$

The second term $\dot{\varphi}\tan\varphi$ is obtained from the camera:

$$\cos\varphi = \tilde{l}^i_z \;\Rightarrow\; \dot{\varphi}\tan\varphi = -\frac{1}{\tilde{l}^i_z}\frac{d}{dt}\tilde{l}^i_z.$$

The inertial-frame derivative of $\tilde{l}^i$ involves the camera-to-inertial rotation applied to the pixel-derived derivative:

$$\frac{d\tilde{l}^i}{dt_i} = \left(\mathcal{R}^i_b\mathcal{R}^b_g\mathcal{R}^g_c\right)\frac{d\tilde{l}^c}{dt_i}, \qquad \frac{d\tilde{l}^c}{dt_i} = Z(\epsilon)\dot{\epsilon} + \tilde{l}^c_{\text{app}}.$$

**TTC formula:**

$$\boxed{t_c = \frac{\mathbb{L}}{\dot{\mathbb{L}}} = \frac{1}{\dfrac{\dot{h}}{h} + \dot{\varphi}\tan\varphi}}$$

Key advantage: no knowledge of object physical size is needed; only the vehicle height $h$ and the pixel/angular-rate measurements.

## 1.2 Precision Landing via Proportional Navigation Guidance

**Problem statement.** Use a camera to guide the UAV to land precisely on a (possibly moving) visually distinct target.

**Navigation model of the UAV:**

$$\dot{p}_n = V_g\cos\chi\cos\gamma, \quad \dot{p}_e = V_g\sin\chi\cos\gamma, \quad \dot{p}_d = -V_g\sin\gamma,$$

$$\dot{\chi} = \frac{g}{V_g}\tan\phi\cos(\chi-\psi), \quad \dot{\phi} = u_1, \quad \dot{\gamma} = u_2.$$

**Proportional Navigation (PN) setup.** Define the LOS vector and its rate:

$$\boldsymbol{l} = p_{\text{obj}} - p_{\text{MAV}}, \qquad \dot{\boldsymbol{l}} = v_{\text{obj}} - v_{\text{MAV}}.$$

PN guidance: maneuver the vehicle so that $\dot{\boldsymbol{l}}$ aligns with $-\boldsymbol{l}$, i.e., bring the relative velocity along the LOS. Guidance command is proportional to

$$\boldsymbol{a}_{\text{MAV}} \propto \boldsymbol{l} \times \dot{\boldsymbol{l}}.$$

When $\dot{\boldsymbol{l}} \parallel \boldsymbol{l}$, this cross-product is zero (target alignment achieved).

**Normalized perpendicular LOS rate.** Since $\dot{\boldsymbol{l}}$ cannot be directly recovered from a monocular camera, define:

$$\boldsymbol{\Omega}_\perp = \tilde{\boldsymbol{l}} \times \frac{\dot{\boldsymbol{l}}}{\mathbb{L}}, \quad \tilde{\boldsymbol{l}} = \frac{\boldsymbol{l}}{\mathbb{L}},$$

and the commanded acceleration (perpendicular to vehicle velocity, in inertial frame):

$$\boxed{\boldsymbol{a}_{\mathrm{MAV}} = N\,\boldsymbol{\Omega}_\perp \times \boldsymbol{v}_{\mathrm{MAV}}, \quad N > 0.}$$

**Camera-frame estimation and frame conversion.**

$$\boldsymbol{\Omega}_\perp^{v2} = \left(\mathcal{R}_b^{v2}\mathcal{R}_g^b\mathcal{R}_c^g\right)\boldsymbol{\Omega}_\perp^c,$$

where $\boldsymbol{\Omega}_\perp^c$ is estimated using pixel data:

$$\frac{\boldsymbol{l}^c}{\mathbb{L}} = \frac{d}{dt}\frac{\boldsymbol{l}^c}{\mathbb{L}} + \frac{\dot{\mathbb{L}}}{\mathbb{L}}\tilde{l}^c = Z(\epsilon)\dot{\epsilon} + \tilde{l}^c_{\mathrm{app}} + \frac{\dot{\mathbb{L}}}{\mathbb{L}}\tilde{l}^c.$$

**Commanded acceleration in vehicle-2 frame:**

$$\boldsymbol{a}_{\mathrm{MAV}}^{v2} = N\boldsymbol{\Omega}_\perp^{v2} \times \boldsymbol{v}_{\mathrm{MAV}}^{v2} = N\begin{bmatrix}\Omega_{\perp,x}^{v2}\\ \Omega_{\perp,y}^{v2}\\ \Omega_{\perp,z}^{v2}\end{bmatrix} \times \begin{bmatrix}V_g\\ 0\\ 0\end{bmatrix} = \begin{bmatrix}0\\ NV_g\Omega_{\perp,z}^{v2}\\ -NV_g\Omega_{\perp,y}^{v2}\end{bmatrix}.$$

**Roll and pitch-rate commands.** General rule:

$$\phi^c = \sigma(a_y^{v2})\tan^{-1}\left(\frac{a_y^{v2}}{|a_z^{v2}|}\right), \quad \dot{\gamma}^c = -\frac{\mathrm{sign}(a_z^{v2})\sqrt{(a_y^{v2})^2 + (a_z^{v2})^2}}{V_g}.$$

The signed-sigmoidal function $\sigma$ handles the discontinuity when $a_z^{v2} = 0$:

$$\sigma(a_y^{v2}) = \mathrm{sign}(a_y^{v2})\frac{1 - e^{-k(a_y^{v2})^2}}{1 + e^{-k(a_y^{v2})^2}}, \quad k > 0.$$

This gives a smooth roll command $\phi^c \in (-\pi/2, \pi/2)$ without discontinuities.

## 1.3 Fixed-wing vs. Multirotor UAVs

| | Fixed-wing | Single-rotor helicopter | Multirotors |
|---|---|---|---|
| Ease-of-use | ++ | + | +++ |
| Reliability | ++ | + | +++ |
| Maintainability | ++ | + | +++ |
| Time of endurance | +++ | ++ | + |
| Payload capacity | ++ | +++ | + |

- Fixed-wing: must maintain forward airspeed; requires runway/launcher; no VTOL.
- Multirotors: VTOL, hover, mechanically simple, high reliability; lower endurance and payload.

### 1.4 Quadrotor Movements (Physical Intuition)

All movements use differential propeller speed changes. Propellers 1–4 in X-config: 1 (front-right, CW), 2 (rear-right, CCW), 3 (rear-left, CW), 4 (front-left, CCW).

**Hover:** All four angular speeds equal; thrusts sum to vehicle weight; reaction torques cancel in pairs.

**Up/Down:** All four $\bar{\omega}_i$ increased/decreased equally $\Rightarrow$ thrust changes, net moment remains zero.

**Forward/Backward:** $\bar{\omega}_{1,4}$ decrease and $\bar{\omega}_{2,3}$ increase by same $\Delta \Rightarrow$ pitch moment forward. Vertical thrust component decreases, so all speeds must be raised additionally to maintain altitude.

**Left/Right:** $\bar{\omega}_{1,2}$ decrease and $\bar{\omega}_{3,4}$ increase $\Rightarrow$ roll right (or vice versa). Same altitude compensation needed.

**Yaw:** $\bar{\omega}_{2,4}$ decrease and $\bar{\omega}_{1,3}$ increase $\Rightarrow$ CW reaction torques dominate $\Rightarrow$ body yaws CW. Zero roll/pitch moment (symmetric change).

# 2 Lecture 19: Quadrotor Hardware and Modelling Architecture

## 2.1 Structural Components

### Landing Gear

- Supports the multicopter on landing/takeoff; keeps propellers above ground.
- Weakens ground effect; absorbs landing impact energy.

### Duct

- Protects blades and ensures personal safety.
- Reduces tip vortex losses $\Rightarrow$ improves thrust efficiency and reduces noise.
- Acts as an inlet–diffuser: generates additional thrust beyond the bare propeller.

### Propeller

- 2-blade: better aerodynamic efficiency.
- 3-blade: smaller diameter for the same thrust, but lower efficiency; can improve endurance if reduction in size/weight dominates.
- *Chord length*: located at 2/3 of the propeller radius (representative chord).
- *Propeller pitch*: distance advanced per revolution in a soft solid.
- *Good match condition*: motor operating point at high efficiency $\Rightarrow$ less power for same thrust.

### Motor and ESC

- BLDC motor converts stored electrical energy (battery) to mechanical energy for propeller.
- ESC (Electronic Speed Controller): receives PWM signal from autopilot, drives BLDC.
- Motor efficiency varies with output torque (type, size, speed).

## 2.2 Airframe Configurations

### Plus (+) vs. X configuration:

- X-configuration is more popular: higher maneuverability (all rotors contribute to pitch and roll), less forward FOV occlusion.
- Arms are at 45° to the forward axis in X-config.

### Ring configuration:

- More rigid, less vibration from motors/propellers.
- Heavier fuselage $\Rightarrow$ lower endurance and slightly reduced maneuverability.

### Propeller forms:

- Common: one motor + propeller per arm.
- Co-axial: two propellers per arm (one top, one bottom). Efficiency of each blade reduced due to airflow interference, but combined thrust $\approx 1.6\times$ single propeller.
- Efficiency requires $h/r_p > 0.357$ (spacing-to-radius ratio).
- Non-horizontal discs: thrust has components along all three body axes $\Rightarrow$ no need to tilt the vehicle for horizontal flight; but reduced vertical thrust capability.
- Variable-pitch propellers: can generate negative thrust (bidirectional).

**Size vs. propeller length:** For $n$-rotor multicopter with arm radius $R$ and max propeller radius $r_{\max}$:

$$\sin\left(\frac{\pi}{n}\right) = \frac{r_{\max}}{R} \;\Rightarrow\; R = \frac{r_{\max}}{\sin(\pi/n)}.$$

Compact design: $r_{\max} = [1.05, 1.2]\, r_p$ where $r_p$ is the nominal propeller radius.

**Minimum propellers for full 6-DOF control:** at least 6 propellers needed.

## 2.3 CG Location

- CG should lie on the central (vertical) axis; off-center CG creates parasitic moments.
- **Forward flight with low CG**: drag torque (force $\times$ CG-arm) makes pitch angle tend to zero — *stable*.
- **Forward flight with high CG**: drag torque increases pitch angle toward flip — *unstable*.
- **Wind gust with high CG**: drag torque self-corrects pitch angle — *stable*.
- **Wind gust with low CG**: drag torque worsens pitch angle — *unstable*.
- **Optimal placement**: as close to center as possible, slightly below the propeller disc plane (balances both scenarios).

## 2.4 Autopilot Placement

- Autopilot carries IMU sensors; ideal position is the geometric center.
- Off-center placement introduces *lever-arm effect*: centrifugal and tangential accelerations contaminate the accelerometer measurement, degrading state estimation.
- Place autopilot near the CG assuming geometric center $\approx$ CG.

## 2.5 Aerodynamic Drag on Fuselage

Four types: frictional drag, pressure drag, induced drag, interference drag. Design goal: smooth joints and streamlined fuselage surfaces.

## 2.6 Modelling Architecture

| Model | Input → Output |
|---|---|
| Rigid-body kinematic model | Velocity, angular velocity → Position, attitude |
| Rigid-body dynamic model | Thrust, moments → Velocity, angular velocity |
| Control effectiveness model | Propeller angular speeds → Thrust, moments |
| Propulsor model | Throttle command → Propeller angular speeds |
| Control allocation model | (inverse of control effectiveness) |

# 3 Lecture 20: Quadrotor Kinematics and Dynamics

## 3.1 Coordinate Frames and Euler Angles

**Inertial frame (NED or ENU)**: fixed, denoted $\mathcal{F}^i$. **Body frame**: $\mathcal{F}^b$ with origin at CoM; $z_b$ pointing *upward* (opposite gravity in hover). Attitude parameterization: Euler angles $\boldsymbol{\Theta} = [\phi, \theta, \psi]^\top$ (roll, pitch, yaw).

**Body-to-inertial rotation matrix** (ZYX convention):

$$\mathcal{R}_b^i = \mathcal{R}_z(\psi)\mathcal{R}_y(\theta)\mathcal{R}_x(\phi).$$

## 3.2 Rigid-Body Kinematics

**Position:**

$$\dot{\boldsymbol{p}}^i = \mathcal{R}_b^i \boldsymbol{v}^b,$$

where $\boldsymbol{p}^i = [p_N, p_E, p_D]^\top$ is position in the inertial frame and $\boldsymbol{v}^b$ is the velocity in the body frame.

**Euler angle kinematics:**

$$\dot{\boldsymbol{\Theta}} = \mathbf{T}(\boldsymbol{\Theta})\,\boldsymbol{\omega}, \quad \mathbf{T}(\boldsymbol{\Theta}) = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix},$$

where $\boldsymbol{\omega} = [p, q, r]^\top$ is the body-frame angular velocity.

*Singularity*: $\mathbf{T}$ is singular at $\theta = \pm\pi/2$ (gimbal lock). Use rotation matrix (SO(3)) formulation for aggressive maneuvers.

## 3.3 Rigid-Body Newton–Euler Dynamics

**Translational dynamics** (in body frame):

$$m\dot{\boldsymbol{v}}^b = \boldsymbol{F}^b - \boldsymbol{\omega} \times (m\boldsymbol{v}^b).$$

In the inertial frame, for quadrotor with $z_b$ up:

$$m\ddot{\boldsymbol{p}} = f\,\mathcal{R}_b^i\,\boldsymbol{e}_3 - mg\,\boldsymbol{e}_3, \quad \boldsymbol{e}_3 = [0, 0, 1]^\top,$$

where $f > 0$ is the total thrust and $\boldsymbol{e}_3$ in the right-hand side is the inertial upward direction.

**Rotational dynamics:**

$$\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) + \boldsymbol{\tau}_{\text{gyro}},$$

where $\mathbf{J} = \text{diag}(J_{xx}, J_{yy}, J_{zz})$ is the diagonal inertia tensor and $\boldsymbol{\tau}$ is the net body-frame torque.

**Gyroscopic torque** due to spinning rotors:

$$\boldsymbol{\tau}_{\text{gyro}} = -J_r\,\boldsymbol{\omega} \times \left(\sum_{k=1}^{n_r} \sigma_k \bar{\omega}_k\right) \hat{\boldsymbol{e}}_3,$$

where $J_r$ is the rotor moment of inertia about its spin axis, $\sigma_k \in \{-1, +1\}$ is the spin direction, and $\bar{\omega}_k$ is angular speed of rotor $k$.

### 3.4 Quadrotor Force and Moment Model

Each propeller $k$ produces:

- Thrust: $f_k = c_T \bar{\omega}_k^2$
- Reaction torque: $\tau_{d,k} = \sigma_k c_M \bar{\omega}_k^2$

where $c_T > 0$ (thrust coefficient), $c_M > 0$ (drag-torque coefficient).

**Total thrust and moments for X-config quadrotor** (arm length $d$, propeller positions at $\psi_i = (i-1)\pi/2 + \pi/4$, $i = 1, 2, 3, 4$; $\sigma = [+1, -1, +1, -1]$):

$$\begin{bmatrix} f_d \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} c_T & c_T & c_T & c_T \\ -\frac{dc_T}{\sqrt{2}} & -\frac{dc_T}{\sqrt{2}} & \frac{dc_T}{\sqrt{2}} & \frac{dc_T}{\sqrt{2}} \\ \frac{dc_T}{\sqrt{2}} & -\frac{dc_T}{\sqrt{2}} & -\frac{dc_T}{\sqrt{2}} & \frac{dc_T}{\sqrt{2}} \\ c_M & -c_M & c_M & -c_M \end{bmatrix}}_{\mathbf{M}_4} \begin{bmatrix} \bar{\omega}_1^2 \\ \bar{\omega}_2^2 \\ \bar{\omega}_3^2 \\ \bar{\omega}_4^2 \end{bmatrix}.$$

The control effectiveness matrix factors as $\mathbf{M}_4 = \mathbf{P}_a \mathbf{M}_4(1,1,1)$ where $\mathbf{P}_a = \mathrm{diag}(c_T, dc_T, dc_T, c_M)$. Since the quadrotor has exactly 4 inputs and 4 outputs, $\mathbf{M}_4$ is square and invertible: $\mathbf{P}_4 = \mathbf{M}_4^{-1}$.

### 3.5 Propulsor Model

The BLDC motor + ESC + propeller system is modelled as a first-order system:

$$\bar{\omega}_k(s) = \frac{1}{T_m s + 1}\big(C_R \sigma_k + \omega_b\big), \quad \sigma_k \in [0, 1],$$

where $T_m$ is the motor time constant, $C_R$ is the speed-range coefficient (rad/s per throttle unit), $\omega_b$ is the baseline (no-load) angular speed, and $\sigma_k$ is the throttle command.

### 3.6 Blade Flapping and Hub Forces

At non-zero forward airspeed, the advancing blade (higher relative airspeed) produces more lift than the retreating blade. This asymmetry causes *blade flapping*: a periodic change in blade pitch/position. Flapping generates *hub forces* (in the plane of the rotor) and cross-coupling pitching/rolling moments. In hover and low-speed flight these are small and often neglected; they become important at high speeds.

# 4 Lecture 21: Quadrotor Position and Velocity Control

## 4.1 Cascaded Control Architecture

The quadrotor control hierarchy (outer to inner):

$$\underbrace{\text{Position loop}}_{\text{Lec 21}} \to \underbrace{\text{Velocity loop}}_{\text{Lec 21}} \to \underbrace{\text{Attitude loop}}_{\text{Lec 22}} \to \underbrace{\text{Motor speed loop}}_{\text{Lec 22}}$$

- Outputs of the position/velocity controllers provide attitude references $(\phi_d, \theta_d)$ and total thrust command $f_d$ to the attitude control system.
- Bandwidth of inner loop must be significantly higher than outer loop for the separation assumption to hold.

## 4.2 Position Dynamics

From Newton's second law with $z_b$ pointing up:

$$\begin{aligned} m\ddot{p}_x &= f(\cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi), \\ m\ddot{p}_y &= f(\sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi), \\ m\ddot{p}_z &= f\cos\theta\cos\phi - mg. \end{aligned}$$

For small roll and pitch angles ($\phi, \theta \ll 1$, $\cos\phi \approx \cos\theta \approx 1$):

$$m\ddot{p}_x \approx f(\cos\psi\,\theta + \sin\psi\,\phi), \quad m\ddot{p}_y \approx f(\sin\psi\,\theta - \cos\psi\,\phi), \quad m\ddot{p}_z \approx f - mg.$$

## 4.3 Altitude (Z) Controller

PID on altitude error $e_z = p_{z,d} - p_z$:

$$f_d = \frac{m}{\cos\theta\cos\phi}\big(g + \ddot{p}_{z,d} + K_{pz}e_z + K_{iz}\int e_z\,dt + K_{dz}\dot{e}_z\big).$$

For small angles: $f_d \approx m\big(g + K_{p,z}e_z + K_{v,z}\dot{e}_z + \cdots\big)$.

Typical cascade: position P loop $\to$ velocity PID loop:

$$\dot{p}_{z,c} = K_{pz}e_z, \qquad f_d = m\big(g + K_{vpz}e_{vz} + K_{viz}\int e_{vz} + K_{vdz}\dot{e}_{vz}\big), \quad e_{vz} = \dot{p}_{z,c} - \dot{p}_z.$$

## 4.4 Horizontal (X–Y) Controller

Desired horizontal accelerations from PID:

$$a_{d,x} = \ddot{p}_{x,d} + K_{px}e_x + K_{ix}\int e_x + K_{dx}\dot{e}_x, \quad a_{d,y} = \ddot{p}_{y,d} + K_{py}e_y + K_{iy}\int e_y + K_{dy}\dot{e}_y.$$

**Desired attitude from desired acceleration.** Given $a_{d,x}, a_{d,y}$, current yaw $\psi$, and thrust $f_d$:

$$\phi_d = \frac{m}{f_d}(a_{d,y}\cos\psi - a_{d,x}\sin\psi), \qquad \theta_d = \frac{m}{f_d}(a_{d,x}\cos\psi + a_{d,y}\sin\psi).$$

(Small-angle approximation; exact versions involve $\sin^{-1}$ and $\tan^{-1}$.) The yaw reference $\psi_d$ is set independently by the mission designer.

Cascade form:

$$\dot{p}_{x,c} = K_{px}e_x, \quad a_{d,x} = K_{vpx}e_{vx} + K_{vix}\int e_{vx} + K_{vdx}\dot{e}_{vx}, \quad e_{vx} = \dot{p}_{x,c} - \dot{p}_x.$$

### 4.5 Attitude Reference Generation

The triple $(\phi_d, \theta_d, \psi_d)$ together with $f_d$ is passed to the attitude controller (Lecture 22). Important: the assumption of fast inner-loop dynamics ($\phi \approx \phi_d$, $\theta \approx \theta_d$) must hold for the position controller to work as designed.

# 5 Lecture 22: Quadrotor Attitude Control, Control Allocation, and Motor Control

## 5.1 Attitude Control Objectives

Design torques $\boldsymbol{\tau}$ such that:

- **Euler angle objective**: $\lim_{t\to\infty} \|\boldsymbol{\Theta}_h(t) - \boldsymbol{\Theta}_{hd}(t)\| = 0$.
- **Rotation matrix objective**: $\lim_{t\to\infty} \|\mathcal{R}^\top \mathcal{R}_d - I_3\| = 0$.

The inner (attitude) loop bandwidth must be significantly higher than the outer (position) loop.

## 5.2 Euler Angle Based Attitude Control

**Error definition:**

$$\boldsymbol{e}_\Theta = \boldsymbol{\Theta} - \boldsymbol{\Theta}_d, \quad \boldsymbol{\Theta}_d = \begin{bmatrix} \boldsymbol{\Theta}_{hd}^\top & \psi_d \end{bmatrix}^\top,$$

where $\boldsymbol{\Theta}_{hd}^\top = [\phi_d, \theta_d]$ comes from the position controller and $\psi_d$ is the designer's choice.

**Desired angular velocity** (from $\dot{\boldsymbol{\Theta}} = \boldsymbol{\omega}$):

$$\boxed{\boldsymbol{\omega}_d = -K_\Theta\, \boldsymbol{e}_\Theta}$$

**Angular velocity error:** $\boldsymbol{e}_\omega = \boldsymbol{\omega} - \boldsymbol{\omega}_d$. With assumption $\dot{\boldsymbol{\Theta}}_d = 0$ and $\boldsymbol{e}_\omega \to 0$: $\boldsymbol{e}_\Theta \to 0$ as $t \to \infty$.

**Torque command (PID, using $\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}$):**

$$\boxed{\boldsymbol{\tau}_d = -K_{\omega p}\boldsymbol{e}_\omega - K_{\omega i}\int \boldsymbol{e}_\omega\, dt - K_{\omega d}\dot{\boldsymbol{e}}_\omega}$$

The derivative term can be omitted to reduce noise sensitivity.

**Saturated form** (with symmetric thrust limit $\tau_{\max}$):

$$\boldsymbol{e}_\omega = \mathrm{sat}_{\mathrm{gd}}(\boldsymbol{\omega} - \boldsymbol{\omega}_d,\, a_{10}), \quad \boldsymbol{\tau}_d = \mathrm{sat}_{\mathrm{gd}}\left(-K_{\omega p}\boldsymbol{e}_\omega - K_{\omega i}\int \boldsymbol{e}_\omega - K_{\omega d}\dot{\boldsymbol{e}}_\omega,\ \tau_{\max}\right) \in [-\tau_{\max}, \tau_{\max}].$$

## 5.3 Rotation Matrix (SO(3)) Based Attitude Control

**Attitude error matrix:**

$$\tilde{\mathcal{R}} = \mathcal{R}^\top \mathcal{R}_d.$$

This is an identity matrix when $\mathcal{R} = \mathcal{R}_d$ (perfect tracking).

**Error vectors:**

$$\boldsymbol{e}_\mathcal{R} = \frac{1}{2}\mathrm{vex}\left(\mathcal{R}_d^\top \mathcal{R} - \mathcal{R}^\top \mathcal{R}_d\right), \qquad \boldsymbol{e}_\omega = \boldsymbol{\omega} - \mathcal{R}^\top \mathcal{R}_d\, \boldsymbol{\omega}_d.$$

**vex**: inverse operator converting a skew-symmetric matrix to a vector. For small angles, $\boldsymbol{\omega}_d \approx \dot{\boldsymbol{\Theta}}_d \approx 0$ so $\boldsymbol{e}_\omega \approx \boldsymbol{\omega}$.

**PD controller** (suitable for slow/moderate maneuvers):

$$\boxed{\boldsymbol{\tau}_d = -K_\mathcal{R}\, \boldsymbol{e}_\mathcal{R} - K_\omega\, \boldsymbol{e}_\omega}$$

**Nonlinear controller** (for aggressive maneuvers, compensates gyroscopic and feedforward terms):

$$\boldsymbol{\tau}_d = -K_{\mathcal{R}}\, \boldsymbol{e}_{\mathcal{R}} - K_\omega\, \boldsymbol{e}_\omega - \mathbf{J}\Big([\boldsymbol{\omega}]_\times \mathcal{R}^\top \mathcal{R}_d\, \boldsymbol{\omega}_d - \mathcal{R}^\top \mathcal{R}_d\, \dot{\boldsymbol{\omega}}_d\Big)\,,$$

where $[\boldsymbol{\omega}]_\times$ is the skew-symmetric (cross-product) matrix of $\boldsymbol{\omega}$.

## 5.4 Control Allocation

**Problem:** Given virtual control input $\boldsymbol{u}_v = [f_d, \boldsymbol{\tau}_d^\top]^\top \in \mathbb{R}^4$, find the propeller speed commands $\bar{\boldsymbol{\omega}}_d^2 = [\bar{\omega}_{d,1}^2, \ldots, \bar{\omega}_{d,n_r}^2]^\top$ satisfying actuator constraints.

**Linear allocation:**

$$\boldsymbol{u}_v(t) = \mathbf{M}_{n_r}\, \bar{\boldsymbol{\omega}}_d^2, \quad \bar{\boldsymbol{\omega}}_d^2 \in \mathbb{R}^{n_r}.$$

Actuator constraints: $\bar{\omega}_{d,k,\min} \le \bar{\omega}_{d,k} \le \bar{\omega}_{d,k,\max}$.

**Quadrotor (unique, $n_r = 4$):**

$$\mathbf{P}_4 = \mathbf{M}_4^{-1}, \qquad \bar{\boldsymbol{\omega}}_d^2 = \mathbf{P}_4 \begin{bmatrix} f_d \\ \boldsymbol{\tau}_d \end{bmatrix}.$$

**Multirotors with $n_r > 4$ (nonunique, uses pseudo-inverse):**

$$\boxed{\mathbf{P}_{n_r} = \mathbf{M}_{n_r}^\dagger = \mathbf{M}_{n_r}^\top \Big(\mathbf{M}_{n_r}\mathbf{M}_{n_r}^\top\Big)^{-1}, \quad \bar{\boldsymbol{\omega}}_d^2 = \mathbf{P}_{n_r} \begin{bmatrix} f_d \\ \boldsymbol{\tau}_d \end{bmatrix}.}$$

**Control effectiveness matrix** for a general $n_r$-rotor vehicle:

$$\mathbf{M}_{n_r}(c_T, c_M, d) = \begin{bmatrix} c_T & c_T & \cdots & c_T \\ -d_1 c_T \sin\varphi_1 & -d_2 c_T \sin\varphi_2 & \cdots & -d_{n_r} c_T \sin\varphi_{n_r} \\ d_1 c_T \cos\varphi_1 & d_2 c_T \cos\varphi_2 & \cdots & d_{n_r} c_T \cos\varphi_{n_r} \\ c_M \delta_1 & c_M \delta_2 & \cdots & c_M \delta_{n_r} \end{bmatrix},$$

where $\varphi_i$ is the azimuth angle of rotor $i$, $d_i$ is its arm length, and $\delta_i \in \{-1, +1\}$ is spin direction.

**Factorization:**

$$\mathbf{M}_{n_r}(c_T, c_M, d) = \mathbf{P}_a\, \mathbf{M}_{n_r}(1,1,1), \quad \mathbf{P}_a = \mathrm{diag}(c_T,\, dc_T,\, dc_T,\, c_M),$$

$$\mathbf{M}_{n_r}^\dagger(c_T, c_M, d) = \mathbf{M}_{n_r}^\dagger(1,1,1)\, \mathbf{P}_a^{-1}.$$

*This factorization means unknown $c_T, c_M, d$ can be compensated via a PID corrective loop on the outputs.*

**Hexacopter ($n_r = 6$, $\varphi_i = (i-1)\cdot 60^\circ$):**

$$\mathbf{M}_6^\dagger(1,1,1) = \frac{1}{6} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 1 & -\sqrt{3} & 1 & -1 \\ 1 & -\sqrt{3} & -1 & 1 \\ 1 & 0 & -2 & -1 \\ 1 & \sqrt{3} & -1 & 1 \\ 1 & \sqrt{3} & 1 & -1 \end{bmatrix}.$$

**Benefits of control allocation:**

- Safer operation: can redistribute thrust even with propeller failure.
- Adapts automatically to different flight conditions.

### 5.5 Motor Speed Control

**Objective:** $\lim_{t\to\infty} |\bar{\omega}_k(t) - \omega_{d,k}(t)| = 0$ for all $k = 1, \ldots, n_r$.

**Propulsor model:**

$$\bar{\omega}_k = \frac{1}{T_m s + 1}(C_R \sigma_k + \omega_b), \quad \sigma_k \in [0, 1].$$

**Closed-loop PID controller** (define error $\tilde{\omega} = \bar{\omega}_k - \omega_{d,k}$):

$$\boxed{\sigma_k = -k_{\omega p}\tilde{\omega} - k_{\omega i}\int \tilde{\omega}\, dt - k_{\omega d}\dot{\tilde{\omega}}_k}$$

**Open-loop (feedforward) controller:**

$$\sigma_k = a\,\omega_{d,k} + b, \quad a, b \in \mathbb{R}.$$

Parameters $a$ and $b$ are determined from the propulsor model: $\sigma_k = (\omega_{d,k} - \omega_b)/C_R$ at steady-state.

### 5.6 Simulation Parameters (Example)

| Parameter | Value | Unit |
|---|---|---|
| $g$ | 9.81 | m/s$^2$ |
| $d$ (arm) | 0.2223 | m |
| $m$ | 1.0230 | kg |
| $c_T$ | $1.4865 \times 10^{-7}$ | N/(RPM$^2$) |
| $c_M$ | $2.9250 \times 10^{-9}$ | N·m/(RPM$^2$) |
| $J_{xx} = J_{yy}$ | 0.0095 | kg·m$^2$ |
| $J_{zz}$ | 0.0186 | kg·m$^2$ |
| $T_m$ | 0.0760 | s |
| $C_R$ | 80.58 | RPM |
| $\omega_b$ | 976.2 | RPM |

Controller gains (X/Y channels: $K_p = 0.5$, $K_{vp} = 1$, $K_{vi} = 0.05$, $K_{vd} = 0.2$; Z channel: $K_p = 4$, $K_{vp} = 2$, $K_{vi} = 4$, $K_{vd} = 0.4$; Roll/Pitch: $K_\Theta = 2$, $K_{\omega p} = 2$, $K_{\omega d} = 0.8$; Yaw: $K_\Theta = 3$, $K_{\omega p} = 3$, $K_{\omega d} = 2.5$).

# 6 Lecture 23: Fully Actuated Unmanned Autonomous Vehicles

## 6.1 Motivation

Conventional multirotors have all propellers pointing in the same (upward) direction. Parallel thrusts collectively counteract gravity but produce *underactuated dynamics*: the vehicle cannot apply an arbitrary wrench (force + torque) in all 6 directions simultaneously. They must tilt to generate horizontal forces.

**Fully actuated (FA) UAVs** extend this by enabling thrust in *arbitrary directions*, making them:

- Capable of decoupled position and attitude control (independently move while maintaining any orientation).
- Suitable as airborne manipulators, inspection drones, etc.

## 6.2 Classification: Fixed-tilt vs. Variable-tilt

**Fixed-tilt:** Propellers have dissimilar but fixed orientations. The control allocation matrix $\mathbf{M}$ is *constant*.

**Variable-tilt:** Propellers are actively tilted using additional servo actuators. $\mathbf{M}$ depends on $N_a$ tilt angles; can switch between underactuated and fully actuated configurations.

## 6.3 Coordinate Frames and Propeller Configuration

**Body frame:** $\Psi_B(o_B, \boldsymbol{x}_B, \boldsymbol{y}_B, \boldsymbol{z}_B)$; origin at CoM; $\boldsymbol{z}_B$ opposite gravity; $\boldsymbol{x}_B$ forward.

**Propeller frame:** $\Psi_{p_i}(o_{p_i}, \boldsymbol{x}_{p_i}, \boldsymbol{y}_{p_i}, \boldsymbol{z}_{p_i})$; origin at rotor CoM; $\boldsymbol{z}_{p_i}$ along thrust direction; $\boldsymbol{x}_{p_i}$ along line $o_B \to o_{p_i}$.

**Per-rotor parameters:**

- Displacement: $\boldsymbol{\xi}_i = o_{p_i}^B$ (position of rotor $i$ in body frame).
- Orientation: $\boldsymbol{u}_i = \mathcal{R}_{p_i}^B(t)\hat{\boldsymbol{z}}$ (thrust direction in body frame).
- For *fixed-tilt*: $\boldsymbol{\xi}_i$ and $\boldsymbol{u}_i$ are constants.
- For *planar multirotor*: $\mathcal{R}_{p_i}^B = \mathcal{R}_z(\psi_i)\mathcal{R}_y(\beta_i)\mathcal{R}_x(\alpha_i)$, where $\psi_i$ is the heading, $\alpha_i$ the cant angle, $\beta_i$ the dihedral angle.

## 6.4 Aerodynamic Wrench and Control-Wrench Matrix

**Aerodynamic wrench** (generalized force = torque + force):

$$\mathbf{W}^i = \begin{bmatrix} \boldsymbol{\tau}^{i\top} & \boldsymbol{f}^{i\top} \end{bmatrix}^\top .$$

**Wrench transformation** between frames:

$$\mathbf{W}^k = \mathbf{A}_i^k \, \mathbf{W}^i, \quad \mathbf{A}_i^k := \begin{bmatrix} \mathcal{R}_i^k & \hat{\boldsymbol{\delta}}_i^k \mathcal{R}_i^k \\ \mathbf{0} & \mathcal{R}_i^k \end{bmatrix},$$

where $\hat{\boldsymbol{\delta}}_i^k$ is the skew-symmetric matrix of the position offset.

**Individual propeller wrench** (in propeller frame $\Psi_{p_i}$, thrust $\lambda_i$, drag-torque-to-thrust ratio $\gamma$, spin direction $\sigma_i \in \{-1, +1\}$):

$$\mathbf{W}_c^{p_i} = \lambda_i \begin{bmatrix} 0 & 0 & \gamma\sigma_i & 0 & 0 & 1 \end{bmatrix}^\top = \lambda_i \begin{bmatrix} \gamma\sigma_i \hat{\boldsymbol{z}} \\ \hat{\boldsymbol{z}} \end{bmatrix}.$$

**Cumulative aerodynamic-control wrench** in body frame:

$$\mathbf{W}_c^B = \sum_{i=1}^{N_p} \mathbf{A}_{p_i}^B \, \mathbf{W}_c^{p_i} \;\Rightarrow\; \begin{bmatrix} \boldsymbol{\tau}_c^B \\ \boldsymbol{f}_c^B \end{bmatrix} = \sum_{i=1}^{N_p} \lambda_i \begin{bmatrix} \gamma\sigma_i \boldsymbol{u}_i + \boldsymbol{\xi}_i \times \boldsymbol{u}_i \\ \boldsymbol{u}_i \end{bmatrix}.$$

**Compact form:**

$$\boxed{\mathbf{W}_c^B = \mathbf{M}\,\boldsymbol{\lambda}, \quad \mathbf{M} \in \mathbb{R}^{6\times N_p},}$$

where $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_{N_p}]^\top$ is the thrust vector.

## 6.5 Actuation Classification

**Underactuated:** $\operatorname{rank}(\mathbf{M}) < 6$.

**Fully actuated:** $\operatorname{rank}(\mathbf{M}) = 6$, $N_p = 6$.

**Overactuated (with redundancy):** $\operatorname{rank}(\mathbf{M}) = 6$, $N_p > 6$.

**Allowable thrust space:**

$$\Lambda = \{\boldsymbol{\lambda} \in \mathbb{R}^{N_p} \mid 0 \le \lambda_i \le \lambda_{\max} \text{ (unidirectional) or } |\lambda_i| \le \lambda_{\max} \text{ (bidirectional)}\}.$$

**Allowable aerodynamic-wrench space:** $\mathcal{W} = \mathbf{M}\Lambda$ (image of $\Lambda$ under $\mathbf{M}$).

**Omnidirectional vehicle:** $\mathcal{W}$ is large enough to produce a wrench that fully counteracts gravity in *any* orientation $\mathcal{R}$ in $\Psi_B$.

## 6.6 Fixed-tilt FA-UAV Designs

### Quad4Hor

- Conventional quadrotor + 4 additional horizontal rotors positioned between the quadrotor arms.
- Added horizontal rotors produce no moment about CoM.
- **Issue**: Horizontal rotors placed below vertical ones cause airflow interference $\Rightarrow$ negative effect on controllability.

### HexC (Hexarotor with Canted Rotors)

- Full actuation by tilting rotors around the axis collinear with their rotor arm.
- All cant angles $\alpha_i$ equal; rotation directions alternate (3 symmetric pairs with opposite spin).
- Large $\alpha_i \Rightarrow$ high horizontal force capability; Small $\alpha_i \Rightarrow$ high flying efficiency.
- Angle bounded by minimum required upward thrust.
- Better disturbance rejection against lateral wind gusts than conventional hexarotor.

**HexCD (Hexarotor with Canted and Dihedral Rotors)**

- Rotors canted and tilted along axis perpendicular to rotor arms ($\alpha_i$, $\beta_i$).
- Direction of dihedral angle $\beta_i$ does not influence UAV performance.
- Maximum $\beta_i$ limited by frame interference; optimal $\beta_i = 0$.
- HexC has greater maneuverability; HexCD has better robustness against rotor failure (nonzero $\beta$).

**CoHexC (Coaxial Hexarotor with Canted Pairs)**

- 12 rotors in 6 coaxial pairs; each pair rotates in opposite directions.
- Drag torques from the two rotors in a pair cancel each other.
- Produces large amount of thrust; fewer airflow interference issues compared to Quad4Hor (both rotors at same speed).

**HexDTet (Double-Tetrahedron Hexarotor)**

- **First attempt** to achieve a hexarotor with noncoplanar rotors.
- 6 rotors on edges of two opposite tetrahedra sharing a common base.
- Elevation angle $\eta_i$: angle between common base and edge containing the rotor; same for all rotors.

**HeptF (Heptarotor with Minimized Frame)**

- One of the first attempts at an omnidirectional UAV with only unidirectional rotors.
- Minimum 7 rotors required.
- No symmetry constraints $\Rightarrow$ all rotors designed to rotate clockwise.
- Airflow interference minimized by imposing a minimum distance between rotors.

**HeptW (Heptarotor with Maximized Wrench)**

- 7 unidirectional rotors on vertices of a regular heptagon in horizontal plane.
- Rotors spin in alternating directions.
- Orientation optimized for maximum omnidirectional wrench generation.

**OctCu (Octarotor Cube)**

- **First multirotor to successfully perform omnidirectional flight.**
- Rotor positions fixed at vertices of a cube.
- All rotors spin in the same direction; symmetric propellers $\Rightarrow$ bidirectional thrust.
- Cage-like frame for safe human-robot interaction.
- Orientation optimized for maximum attainable omnidirectional wrench (vehicle agility).

**OctB (Octarotor Beam)**

- Omnidirectional octarotor with a beam-like shape; designed for aerial physical-interaction tasks.
- Each side of beam: 4 rotors at square vertices (1,6,7,8 CW; others CCW).
- Bidirectional thrust: two unidirectional propellers stacked in opposite directions on same shaft.

## 6.7 Variable-tilt FA-UAV Designs

**QuadvC (Quadrotor with Variable Cant Rotors)**

- Full actuation via 4 tilting actuators, each canting a rotor individually around $\boldsymbol{x}_{p_i}$-axis.
- Unidirectional rotors; max achievable hover pitch/roll angle $\approx 30°$.

**QuadvD (Quadrotor with Variable Dihedral Rotors)**

- Active tilt of dihedral angle (not cant angle) by extra actuator.
- Inward dihedral limited by rotor blade-frame interference.
- Dihedral does not contribute to yaw torque $\Rightarrow$ reduced total torque.

**QuadvCD (Quadrotor with Variable Cant and Dihedral)**

- Combines both QuadvC and QuadvD: two actuators per rotor.
- 8 total actuators for thrust vectoring.
- Push-pull mechanism limits maximum achievable dihedral angle.

**QuadvCDc (Coupled Cant and Dihedral)**

- Single parallelogram linkage actuates all propellers in the same direction simultaneously.
- Reduces actuators from 8 to 2.
- **Drawback**: Cannot orient rotors individually; limited maximum tilting angle.

**HexvC (Hexarotor with Variable Cant)**

- **First variable-tilt concept to achieve successful omnidirectional flight.**
- Canting actuator on each rotor of a conventional hexarotor.
- Developed at ETH Zurich and Zurich University of the Arts.

**HexvCc (Hexarotor with Coupled Variable Cant)**

- Single actuator to cant all 6 hexarotor rotors simultaneously via wire mechanism.
- Cant angle alternates (same pattern as HexC); maximum cant limited by frame.

- **Advantage**: 6 actuators → 1 (lower mass, less energy).
- **Disadvantage**: Reduced maneuverability; lower bandwidth from wire linkage.

### 6.8 Summary Table: FA-UAV Configurations

| Concept | Abbrev. | Key feature |
|---|---|---|
| Quadrotor + 4 horizontal rotors | Quad4Hor | Simple fixed-tilt full actuation |
| Hexarotor with canted rotors | HexC | Best maneuverability in fixed-tilt hex |
| Hexarotor canted + dihedral | HexCD | Rotor-failure robustness |
| Coaxial hexarotor canted | CoHexC | High thrust, 12 rotors |
| Double-tetrahedron hexarotor | HexDTet | First noncoplanar hex |
| Heptarotor minimized frame | HeptF | First 7-rotor omnidirectional |
| Heptarotor maximized wrench | HeptW | Optimized omni wrench |
| Octarotor cube | OctCu | First omni flight achieved |
| Octarotor beam | OctB | Physical interaction tasks |
| Quad variable cant | QuadvC | $\pm 30°$ tilt capability |
| Quad variable dihedral | QuadvD | Dihedral tilt |
| Quad variable cant+dihedral | QuadvCD | Full individual tilt, 8 actuators |
| Quad coupled cant+dihedral | QuadvCDc | 2 actuators via linkage |
| Hex variable cant | HexvC | First successful omni variable-tilt |
| Hex coupled variable cant | HexvCc | Wire-coupled, 1 actuator |

# 7 Consolidated Revision Checklist (Lectures 18–23)

1. Derive TTC using the flat-earth model from camera pixel rates and vehicle altitude.
2. Write the PNG guidance law, define $\mathbf{\Omega}_\perp$, and convert it to roll/pitch-rate commands using the vehicle-2 frame.
3. Explain the signed-sigmoidal function used to resolve the roll-angle discontinuity.
4. Compare fixed-wing, helicopter, and multirotor UAVs across endurance, payload, reliability.
5. Describe quadrotor movements (up, forward, lateral, yaw) in terms of propeller speed changes.
6. Explain the role of each hardware component: landing gear, duct, motor+ESC, propeller blade count, airframe config.
7. State the optimal CG placement for forward flight vs. wind-disturbance stability and explain the physics.
8. Write out the 4-layer modelling architecture and state the input/output of each layer.
9. Derive the quadrotor Newton–Euler equations of motion including gyroscopic torque.
10. Write the control effectiveness matrix $\mathbf{M}_4$ for an X-config quadrotor from first principles.
11. Explain the factorization $\mathbf{M}_{n_r} = \mathbf{P}_a \mathbf{M}_{n_r}(1,1,1)$ and how it handles unknown coefficients.
12. Describe the cascade position–velocity–attitude control architecture; derive desired $\phi_d, \theta_d$ from desired acceleration.
13. Compare Euler angle and SO(3) rotation matrix attitude controllers: error definitions, controllers, when to use each.
14. Write the saturated PID attitude torque command using $\mathrm{sat}_{\mathrm{gd}}$.
15. Derive the nonlinear SO(3) controller including gyroscopic and feedforward terms.
16. Explain control allocation: pseudo-inverse for overactuated systems, benefits, procedure.
17. Write closed-loop and open-loop motor speed controllers from the propulsor model.
18. Define aerodynamic wrench and write the compact form $\mathbf{W}_c^B = \mathbf{M}\boldsymbol{\lambda}$.
19. Distinguish underactuated, fully actuated, and overactuated configurations using $\mathrm{rank}(\mathbf{M})$.
20. Explain omnidirectionality: what $\mathcal{W}$ must satisfy and why it requires $\mathrm{rank}(\mathbf{M}) = 6$.
21. For at least 5 FA-UAV designs, state the key feature, trade-off, and actuation type.
22. Compare fixed-tilt and variable-tilt concepts: trade-offs in efficiency, maneuverability, and actuator count.

## Primary Reference

Randal Beard and Timothy W. McLain, *Small Unmanned Aircraft: Theory and Practice*, Princeton University Press, 2012.